Practice with continous variables in R, Histograms, Box Plots, and Density Plots

1 Continuous Variables in R

In this notes, we will practice with continuous variables in R, focusing on histograms, box plots, and density plots. We will start by loading the necessary libraries and the "diamonds" dataset.

```
library("tidyverse")
library("here")
library("patchwork")
library("krulRutils")

options(scipen = 999) # Disable scientific notation
data(diamonds)
```

Before starting plotting our data, we first take a little glimpse at the data contained in this dataset:

```
diamonds %>%
  glimpse()
```

```
Rows: 53,940
Columns: 10
$ carat   <dbl> 0.23, 0.21, 0.23, 0.29, 0.31, 0.24, 0.24, 0.26, 0.22, 0.23, 0.~
$ cut     <ord> Ideal, Premium, Good, Premium, Good, Very Good, Very Good, Ver~
$ color   <ord> E, E, E, I, J, J, I, H, E, H, J, J, F, J, E, E, I, J, J, J, I,~
$ clarity <ord> SI2, SI1, VS1, VS2, SI2, VVS2, VVS1, SI1, VS2, VS1, SI1, VS1, ~
$ depth   <dbl> 61.5, 59.8, 56.9, 62.4, 63.3, 62.8, 62.3, 61.9, 65.1, 59.4, 64~
$ table   <dbl> 55, 61, 65, 58, 58, 57, 57, 55, 61, 61, 55, 56, 61, 54, 62, 58~
```

```
$ price   <int> 326, 326, 327, 334, 335, 336, 336, 337, 337, 338, 339, 340, 34~
$ x       <dbl> 3.95, 3.89, 4.05, 4.20, 4.34, 3.94, 3.95, 4.07, 3.87, 4.00, 4.~
$ y       <dbl> 3.98, 3.84, 4.07, 4.23, 4.35, 3.96, 3.98, 4.11, 3.78, 4.05, 4.~
$ z       <dbl> 2.43, 2.31, 2.31, 2.63, 2.75, 2.48, 2.47, 2.53, 2.49, 2.39, 2.~
```

We also look at the most important summary statistics:

```
diamonds %>%
  summarise_numeric_tidy()
```

```
# A tibble: 7 x 8
  statistic carat depth table       price     x     y      z
  <fct>     <dbl> <dbl> <dbl>       <dbl> <dbl> <dbl>  <dbl>
1 min       0.2   43    43            326  0     0      0
2 q1        0.4   61    56            950  4.71  4.72   2.91
3 median    0.7   61.8  57           2401  5.7   5.71   3.53
4 q3        1.04  62.5  59           5324. 6.54  6.54   4.04
5 max       5.01  79    95          18823 10.7  58.9   31.8
6 mean      0.798 61.7  57.5         3933. 5.73  5.73   3.54
7 var       0.225  2.05  4.99 15915629.  1.26  1.30   0.498
```

1.1 Box Plots

Box plots are a great way to visualize the distribution of continuous variables. The following figure shows the main components of a box plot:

```
boxplot_explanation_plot <- ggplot_box_legend()
ggsave(
  filename = here("images", "boxplot_explanation.png"),
  plot = boxplot_explanation_plot,
  width = 6,
  height = 12,
  dpi = 300
)
boxplot_explanation_plot
```

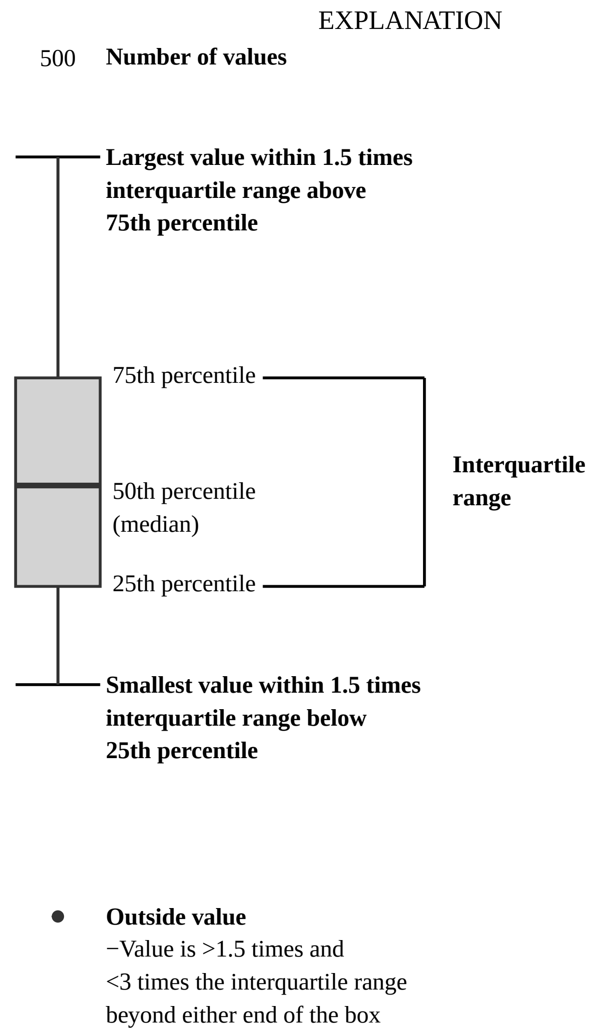

Figure 1: Description of the main components of a boxplot

Now we will construct the boxplot associated to the "price" variable of this dataset:

```
diamonds_boxplot <- diamonds %>%
  ggplot(aes(x = price, y = 1)) +
  geom_boxplot(
    color = "black",
    fill = c_pal("C cyan"),
    outlier.shape = NA
  ) +
  stat_boxplot(geom = "errorbar", width = 0.3, linewidth = 0.5) +
  geom_jitter(height = 0.3, alpha = 0.1, size = 0.1, color = c_pal("C red")) +
```

```r
  coord_cartesian(xlim = c(0, 19000)) +
  labs(
    title = "Diamonds price boxplot",
    x = "Price (USD)",
    y = ""
  ) +
  theme(axis.ticks.y = element_blank(), axis.text.y = element_blank()) +
  theme_krul()

diamonds_boxplot
```

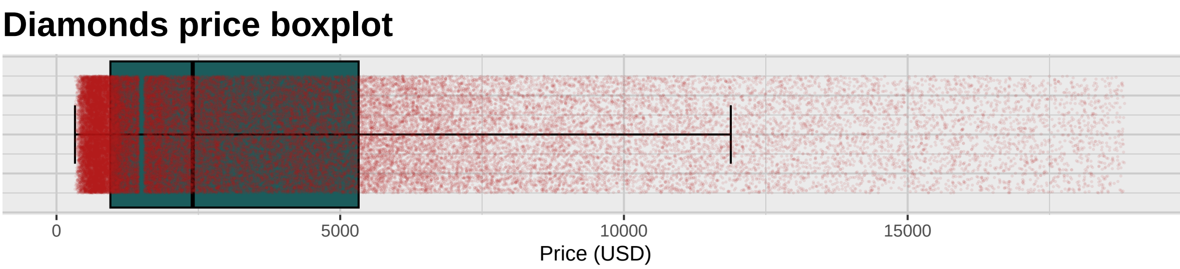

Figure 2: Boxplot showing the general distribution of diamond prices

Now we will do a more detailed analysis by comparing the boxplots corresponding to the different categories associated to the variable "cut".

```r
diamonds_boxplot_by_cut <- diamonds %>%
  ggplot(aes(x = price, y = 1, fill = cut)) +
  geom_boxplot(
    color = "black",
    outlier.shape = NA
  ) +
  stat_boxplot(geom = "errorbar", width = 0.3, linewidth = 0.5) +
  geom_jitter(
    aes(color = cut),
    width = 0.375,
    alpha = 0.1,
    size = 0.1
  ) +
  facet_wrap(~cut, ncol = 1) +
  c_scale_fill("C spring green", "C cyan", "C azure", "C blue", "C violet") +
  c_scale_color("C rose", "C red", "C orange", "C yellow", "C chartreuse") +
  coord_cartesian(xlim = c(0, 19000)) +
```

```
  labs(
    title = "Diamonds price by cut boxplot",
    x = "Price (USD)",
    y = "",
  ) +
  theme(
    axis.ticks.y = element_blank(),
    axis.text.y = element_blank(),
    legend.position = "none"
  ) +
  theme_krul()

diamonds_boxplot_by_cut
```

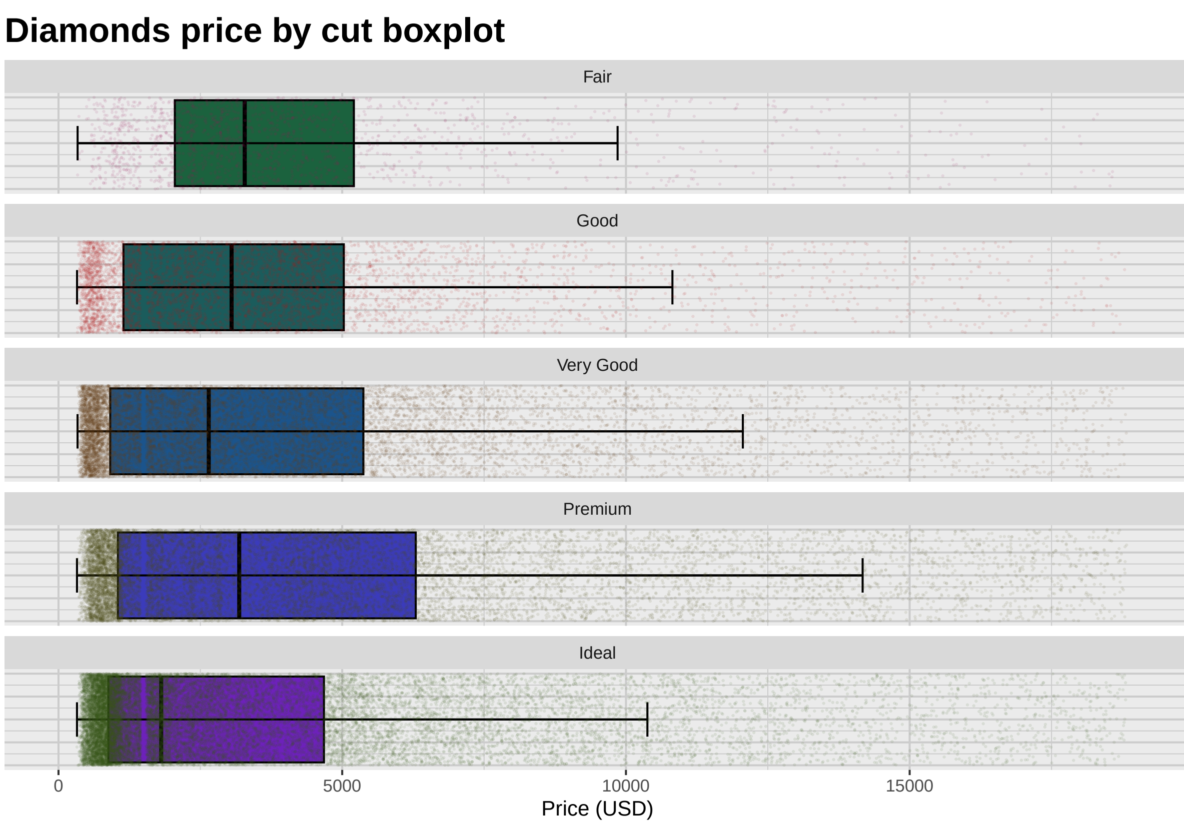

Figure 3: Boxplot showing the distribution of diamond prices by 'cut'

1.2 Histograms

We will now consider the histograms associated with this data. As before we start by looking at the histogram associated to the variable "price":

```
diamonds_histogram <- diamonds %>%
  ggplot(aes(x = price)) +
  geom_histogram(binwidth = 500, color = "black", fill = c_pal("C cyan")) +
  labs(
    title = "Diamonds price histogram",
    x = "Price (USD)",
    y = "Count"
  ) +
  coord_cartesian(xlim = c(0, 19000)) +
  theme_krul()

diamonds_histogram
```

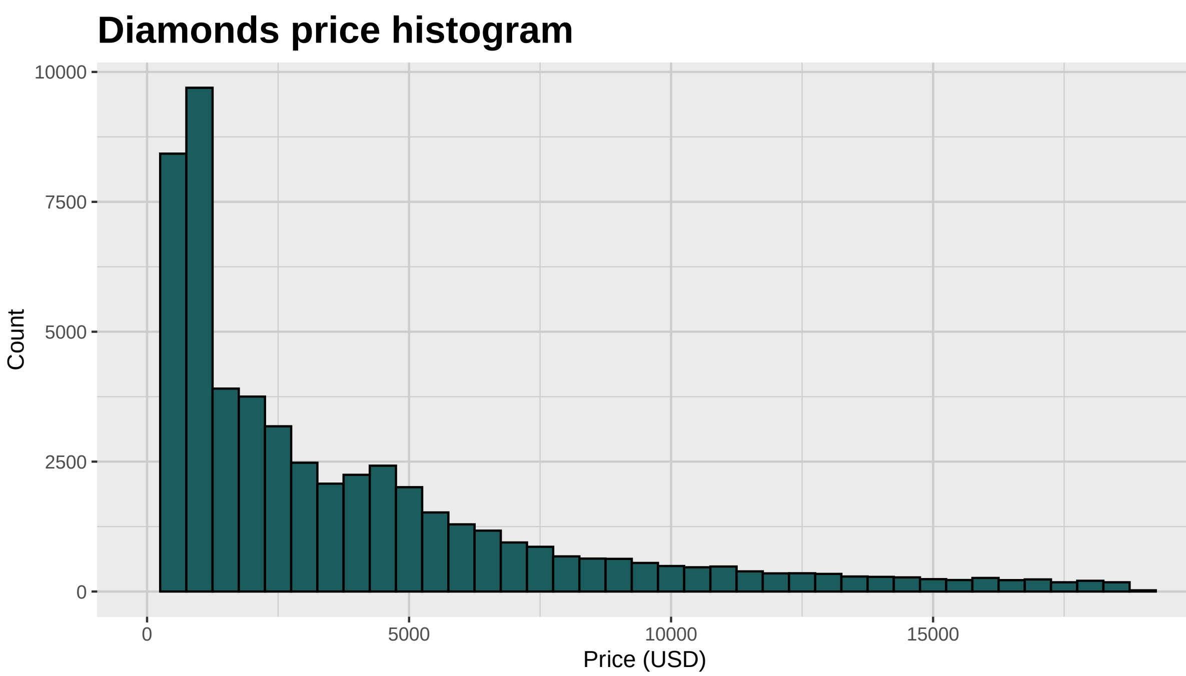

Figure 4: Histogram corresponding to the variable price

We no proceed to a more detailed analysis by breaking our plot according to the different categories associated to the variable "cut"

```r
diamonds_histogram_by_cut <- diamonds %>%
  ggplot(aes(x = price, fill = cut)) +
  geom_histogram(binwidth = 500, color = "black") +
  facet_wrap(~cut, ncol = 1, scales = "free_y") +
  c_scale_fill("C spring green", "C cyan", "C azure", "C blue", "C violet") +
  labs(
    title = "Histogram of the price variable by cut",
    x = "Price (USD)",
    y = "Count"
  ) +
  coord_cartesian(xlim = c(0, 19000)) +
  theme_krul() +
  theme(legend.position = "none")

diamonds_histogram_by_cut
```

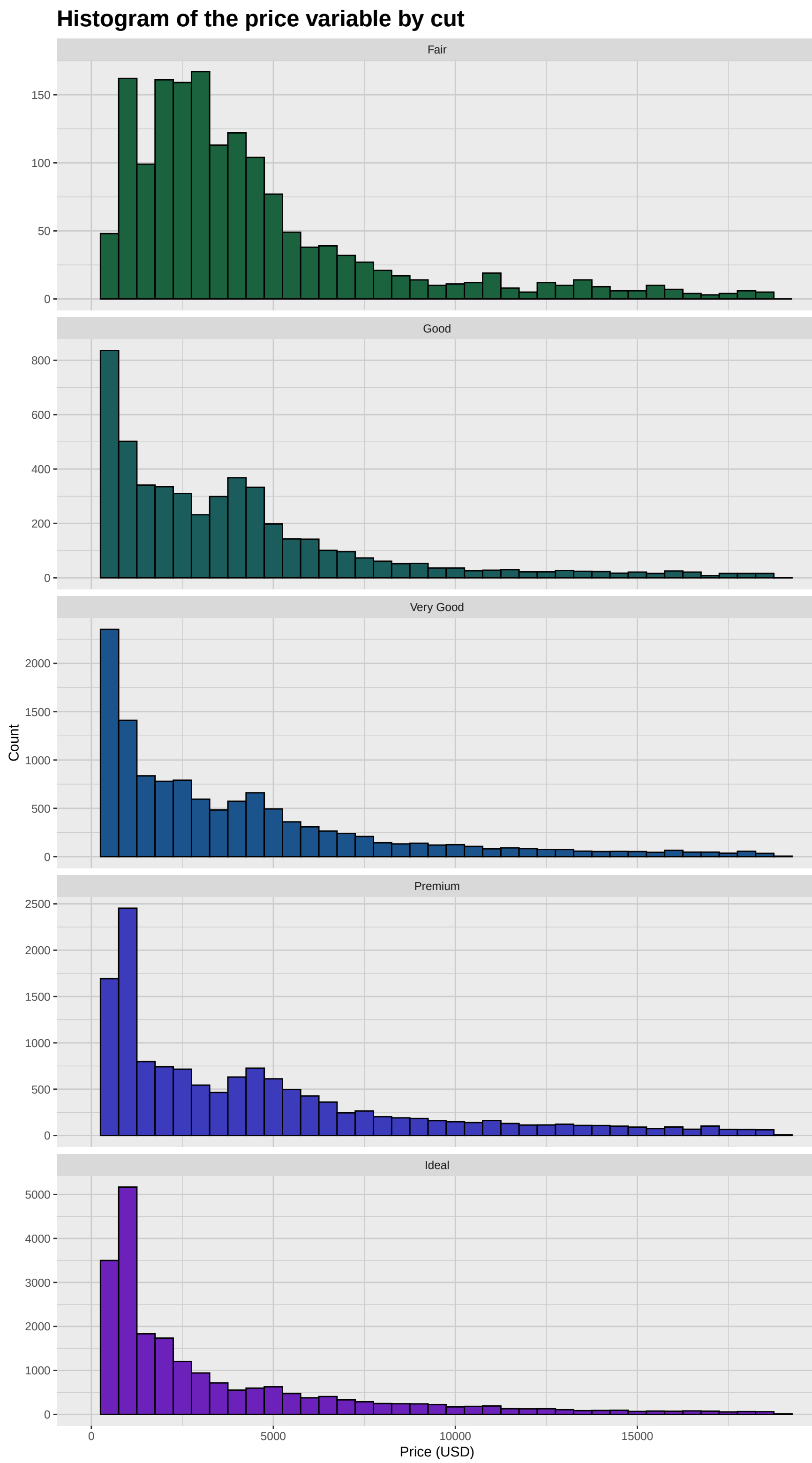

Figure 5: Histogram corresponding to the variable price divided according to the variable 'cut'

1.3 Density plots

Finally, we will analyze the density plots associated to the "price" variable. Density plots provide similar information to histograms, but instead of showing the actual counts, it tries to estimate the shape of the associated probability density funtion.

```
diamonds_density_plot <- diamonds %>%
  ggplot(aes(x = price)) +
  geom_density(color = "black", fill = c_pal("C cyan")) +
  labs(
    title = "Diamonds density plot",
    x = "Price (USD)",
    y = "Density"
  ) +
  coord_cartesian(xlim = c(0, 19000)) +
  theme_krul()

diamonds_density_plot
```

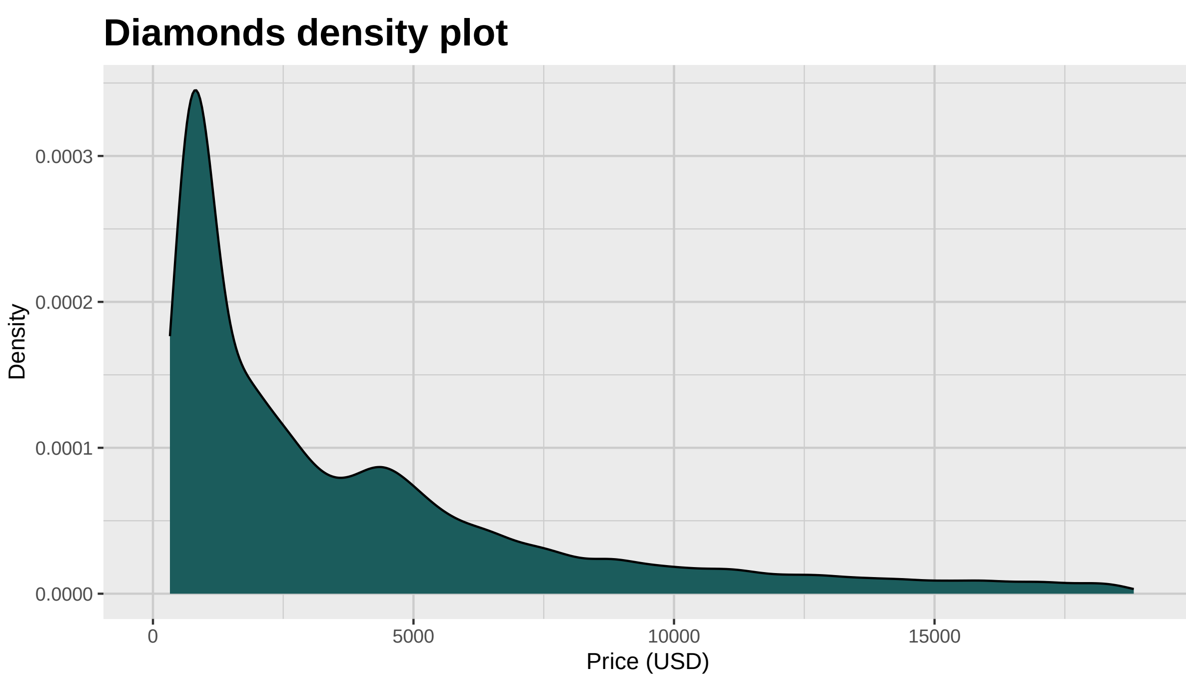

Figure 6: Density plot corresponding to the variable price

As before, we will now perform a more detailed analysis taking the "cut" variable into account

```r
diamonds_density_plot_by_cut <- diamonds %>%
  ggplot(aes(x = price, fill = cut)) +
  geom_density(color = "black") +
  coord_cartesian(xlim = c(0, 19000)) +
  facet_wrap(~cut, ncol = 1, scale = "free_y") +
  c_scale_fill("C spring green", "C cyan", "C azure", "C blue", "C violet") +
  labs(
    title = "Diamonds density plot by cut",
    x = "Price (USD)",
    y = "Density"
  ) +
  theme_krul() +
  theme(legend.position = "none")

diamonds_density_plot_by_cut
```

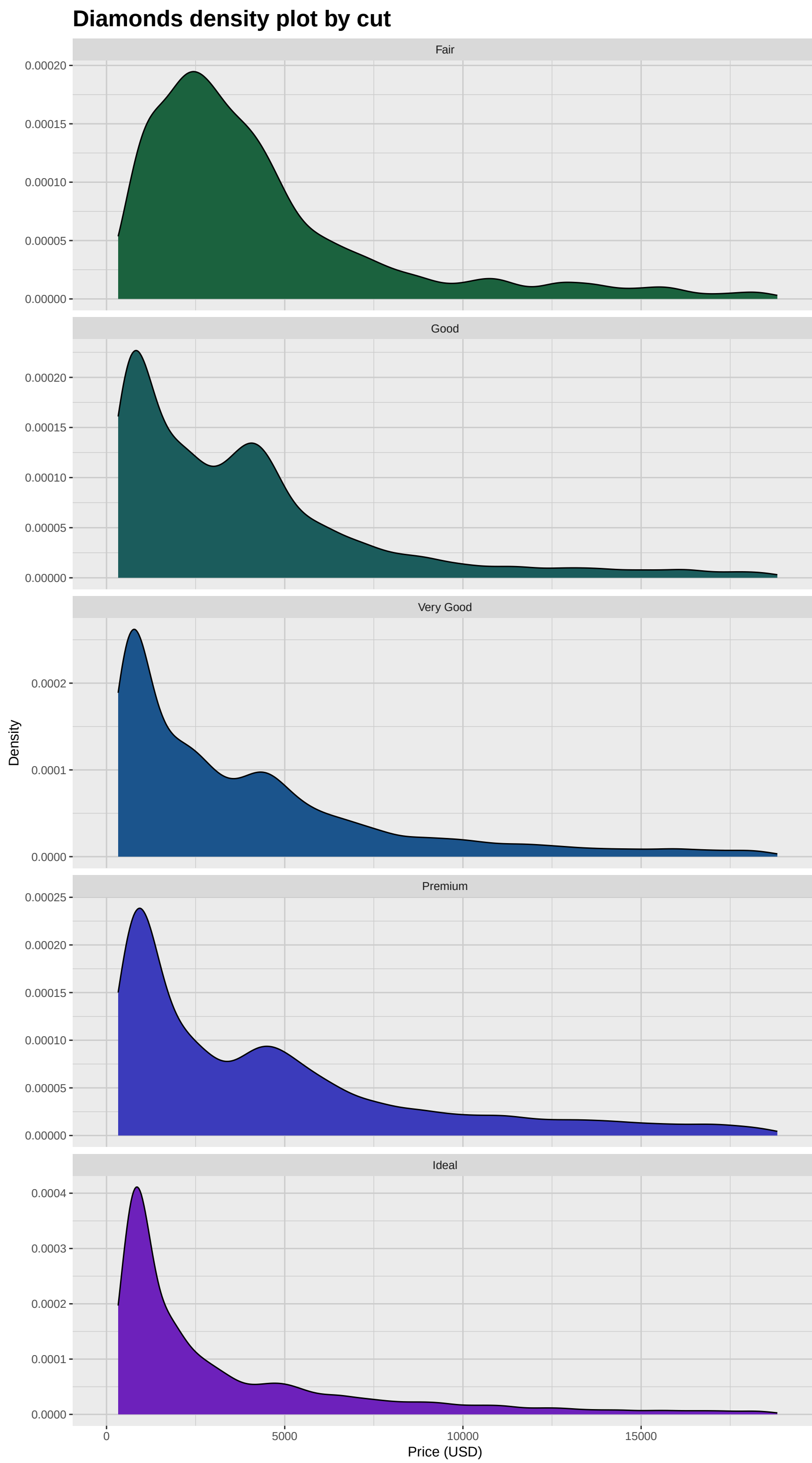

Figure 7: Density plot corresponding to the variable price divided according to the variable ‘cut’

Density plots associated to different categories are frequently plot together. In out case, the 5 categories make it hard to distinguish the plots when they are all shown together, but we can filter the categories to illustrate this option.

```
diamonds_filtered_density_plot <- diamonds %>%
  filter(cut %in% c("Fair", "Ideal")) %>%
  ggplot(aes(x = price, fill = cut)) +
  geom_density(color = "black", alpha = 0.7) +
  coord_cartesian(xlim = c(0, 19000)) +
  c_scale_fill("C spring green", "C violet") +
  labs(
    title = "Diamonds density plot by filtered cut",
    x = "Price (USD)",
    y = "Density"
  ) +
  theme_krul() +
  theme(legend.position = "none")

diamonds_filtered_density_plot
```

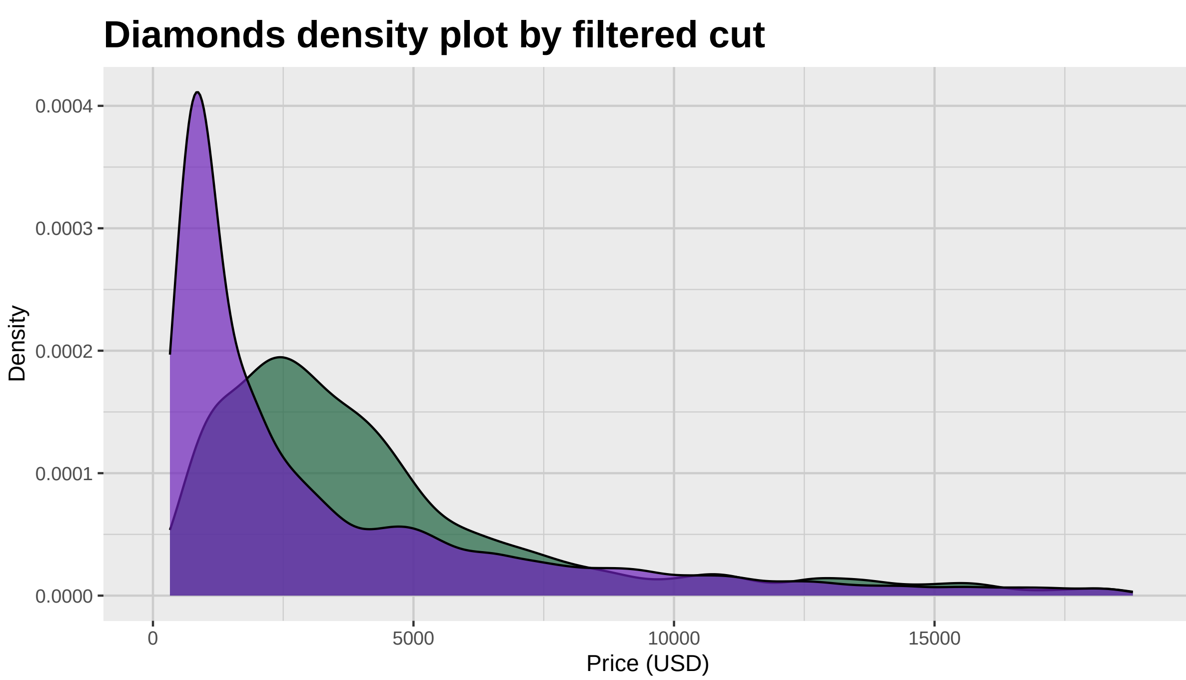

Figure 8: Density plot corresponding to the variable price divided according to the filtered variable 'cut'

2 Using patchwork to put plots together

Finally in this section we will show how we can use patchwork to put all of this information together. Since we have already covered how to find all of these plots, we will just show the corresponding code.

```
diamonds_density_plot <-
  diamonds %>%
  ggplot(aes(x = price, fill = cut)) +
  geom_density(
    alpha = 1,
    color = "black"
  ) +
  c_scale_fill("C spring green", "C cyan", "C azure", "C blue", "C violet") +
  facet_wrap(~cut, nrow = 1, scales = "free_y") + # facet by columns
  coord_cartesian(xlim = c(326, 18823)) +
  labs(
    title = "Density Plots",
    x = "Price (USD)",
    y = "Density",
    fill = ""
  ) +
  theme_krul() +
  theme(strip.background = element_blank(), strip.text = element_blank())

diamonds_histogram <- diamonds %>%
  ggplot(aes(x = price, fill = cut)) +
  geom_histogram(binwidth = 500, color = "black") +
  coord_cartesian(xlim = c(326, 18823)) +
  facet_wrap(~cut, nrow = 1, scales = "free_y") + # facet by columns
  c_scale_fill("C spring green", "C cyan", "C azure", "C blue", "C violet") +
  labs(
    title = "Histograms",
    x = "Price (USD)",
    y = "Count",
    fill = "Cut"
  ) +
  theme_krul() +
  theme(strip.background = element_blank(), strip.text = element_blank())

diamonds_box_plot <- diamonds %>%
  ggplot(aes(x = price, y = 1, fill = cut)) +
```

```r
  geom_boxplot(
    color = "black",
    outlier.shape = NA
  ) +
  geom_jitter(
    aes(color = cut),
    width = 0.375,
    alpha = 0.1,
    size = 0.1
  ) +
  facet_wrap(~cut, nrow = 1) +
  c_scale_fill("C spring green", "C cyan", "C azure", "C blue", "C violet") +
  c_scale_color("C rose", "C red", "C orange", "C yellow", "C chartreuse") +
  coord_cartesian(xlim = c(326, 18823)) +
  labs(
    title = "Box Plots",
    x = "Price (USD)",
    y = "",
    fill = "Cut"
  ) +
  theme_krul() +
  theme(strip.background = element_blank(), strip.text = element_blank()) +
  theme(axis.ticks.y = element_blank(), axis.text.y = element_blank()) +
  stat_boxplot(geom = "errorbar", width = 0.3, size = 0.5)
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```

```r
diamonds_density_plot <- diamonds_density_plot +
  theme(legend.position = "bottom")

diamonds_histogram <- diamonds_histogram +
  theme(legend.position = "none")

diamonds_box_plot <- diamonds_box_plot +
  theme(legend.position = "none")


# Now stack vertically (rows) using patchwork's `/`
diamonds_combined_plot <- (
  diamonds_box_plot /
```

```
    diamonds_histogram /
    diamonds_density_plot
) +
  plot_layout(heights = c(1, 3, 3)) +
  plot_annotation(
    title = "Distribution of Diamond Prices by Cut",
    theme = theme(
      plot.title =
        element_text(
          hjust = 0.5,
          size = 24,
          face = "bold"
        )
    )
  )

ggsave(
  filename = here("images", "diamonds_combined_plot.png"),
  plot = diamonds_combined_plot,
  width = 16,
  height = 9,
  dpi = 300
)

diamonds_combined_plot
```

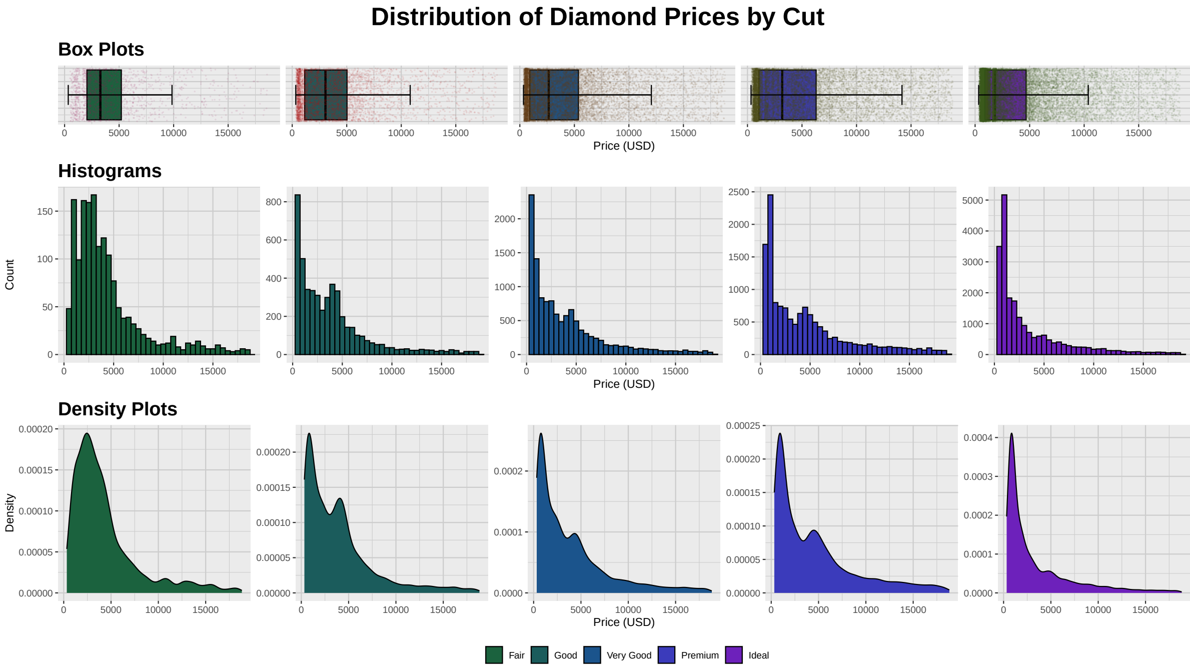

Figure 9: Plot showing the boxplots, histograms and density plots corresponding to the price of diamonds by cut